up front—Elasticsearch is forgiving on that count. We can index a document without letting the engine know anything about the fields' data types. Consider a movie document consisting of a few fields, shown in figure 4.3.

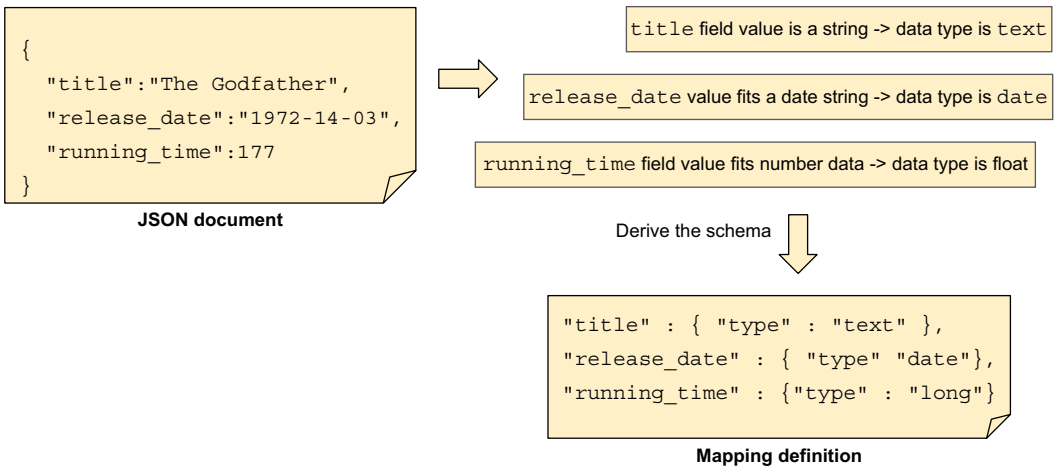


Figure 4.3 Dynamically deriving the indexing schema

Reading the fields and values in this document, Elasticsearch infers the data types automatically:

- The `title` field value is a string, so Elasticsearch maps the field to a `text` data type.
- The `release_date` field value is a string, but the value matches an ISO 8601 date format, so it is mapped to a `date` type.
- The `rating` field is a number, so Elasticsearch maps it to a `float` data type.

Because we do not explicitly create the mapping but instead let Elasticsearch derive the schema on the fly, this type of mapping is called *dynamic mapping*. This feature is handy during application development or in testing environments.

How does Elasticsearch know the data type of the `rating` field is a `float` type? For that matter, how does Elasticsearch deduce any of the types? Let's look next at the mechanism and rules of how types are derived by Elasticsearch.

4.2.1 The mechanism for deducing types

Elasticsearch analyzes field values and guesses the equivalent data type based on those values. When the incoming JSON object is parsed, the `rating` value of 4.9, for example, resembles a floating-point integer in programming language lingo. This "guesswork" is pretty straightforward; hence, Elasticsearch stamps the `rating` field as a `float` field.